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(CmlC)
18 August 1961
SUPERSEDING
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MILITARY SPECIFICATION

SCREEN, FLOCKED GRID

1. SCOPE

1.1 This specification covers one type of flock-covered plastic-grid screen, covered on one side nonly.

2. APPLICABLE DOCUMENTS

2.1 *Government documents.* The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-S-137	- Screening, Plastic Coated Fibrous Glass, Insect.
UU-P-268	- Paper, Kraft, Wrapping
UU-P-271	- Paper, Wrapping, Waterproofed Kraft
UU-T-111	- Tape, Paper, Gummed (Sealing and Securing)
CCC-T-191	- Textile Test Methods
PPP-B-636	- Box, Fiberboard
PPP-T-76	- Tape; Pressure-Sensitive, Adhesive, Paper, Water-Resistant

MILITARY

MIL-A-140	- Adhesive, Water-Resistant, Waterproof Barrier-Material

FSC 4240

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STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Base screen The base screen shall be plastic-coated fibrous glass insect screening, 18 by 14 mesh, aluminum color, conforming to Specification L-S-137. The base screen shall be mildew resistant.

3.1.2 Flock. The flock which is to be applied to the base screen (see 3.1.1) shall meet the following requirements:

(a) Fibers shall be 100 percent virgin viscose with a dull luster.
(b) Fibers shall be 3 denier.
(c) Nominal fiber length shall be 0.030 inch long. Eighty percent of the fibers shall be between 0.027 inch and 0.032 inch long.
(d) Fibers shall be cut squarely with no more than 5.0 percent showing shattered and hooked ends.
(e) Fibers shall be straight, and no more than 15 percent shall be curved in excess of 0.002 inch.

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(f) Fibers shall be dispersible and have good activity in an electrostatic field.

(g) Fibers shall be mildew resistant.

(h) The fibers shall be dyed color-fast black before application to the base screen.

(i) The fibers shall have a moisture content of 11.0 ± 0.5 percent.

3.1.3 Adhesive. The adhesive shall have characteristics such that when used in conjunction with the base screen and flock fibers specified in 3.1.1 and 3.1.2 the flocked grid screen shall meet all requirements of this specification. The adhesive shall not be detrimental to the base screen or flock fibers (see 6.3). The adhesive shall be mildew resistant.

3.2 Flocked grid screen. The flock fibers shall be uniformly deposited on one side of the base screen. Orientation of the fibers and approximate density shall be as shown in figure 1. There shall be no unbonded fibers on the finished flocked grid screen (blowing or sucking the loose fibers from the flocked screen is permitted). There shall be no more than six plugged holes per examined circle (see figure 2) when examined as specified in 4.5.2.

3.2.1 Air permeability.

3.2.1.1 Dry. The air permeability of the dry flocked grid screen shall be no less than 300 and no greater than 500 cubic feet per minute when tested as specified in 4.6.1.

3.2.1.2 Wet. When the flocked grid screen is treated as specified in 4.6.1.1, the air permeability shall be no less than 10.0 percent of the dry screen (4.6.1).

3.2.2 Adhesion of flock. WThere shall be no flock-free base screens before and after treatments specified in table I when tested in accordance with paragraph 4.6.2.

Table I. Treatments

Treatment	Method
Immersion in boiling water	4.6.3
Accelerated weathering	4.6.4
Low temperature treatment	4.6.5
Resistance to mildew	4.6.6

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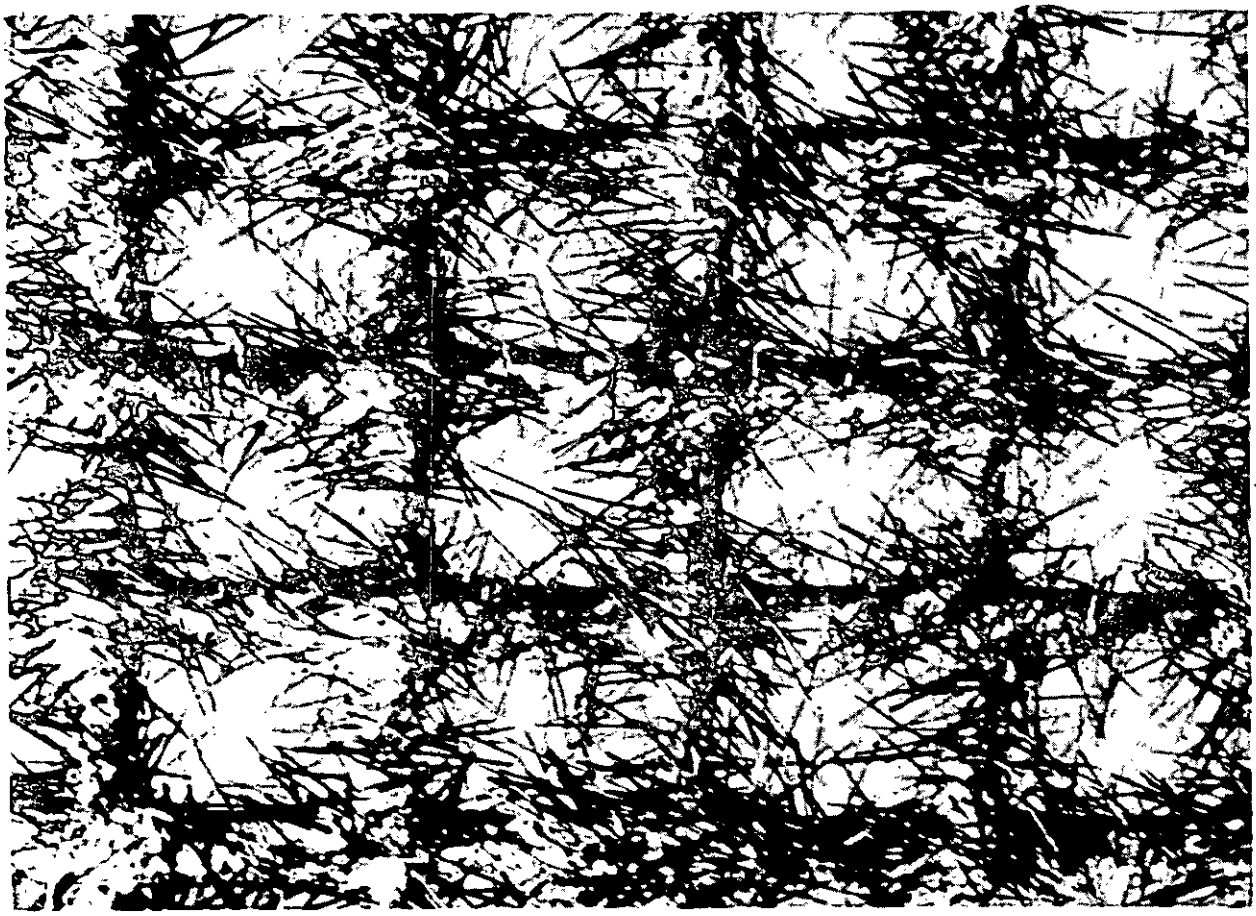


FIGURE 1

Screen,. Flocked Grid
Magnified 20 X

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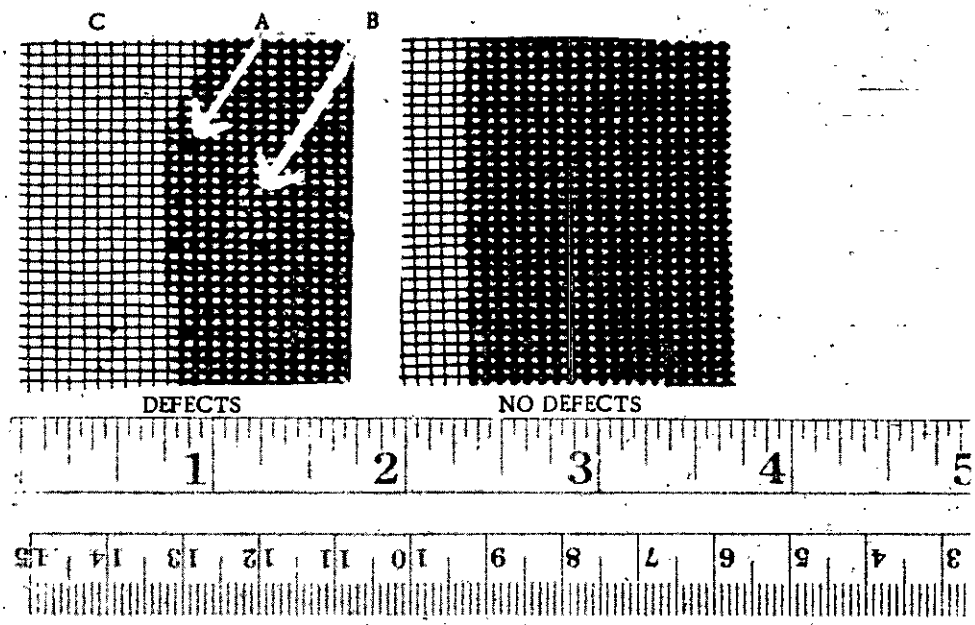


FIGURE 2

Visual Defects

A - Plugged hole
B - Flock free
C - Edge

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3.2.3 Blocking. There shall be no blocking of the flocked grid screen when tested as specified in 4.6.7.

3.2.4 Dimensions. The dimensions of the flocked grid screen shall be specified by the contract or order (see 6.2).

3.2.4.1 Selvedge edge. No more than 3/4 inch of the edge shall be free of the flock covering (see figure 2) when measured in three places on each edge.

3.3 Workmanship. The flock-covered grid screen shall be free from contamination (foregin material) and imperfections which would impair its appearance or serviceability, such as nicks, scrapes, scratched, or tears.

4. QUALITY ASSURANCES PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection factilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirement of 3.1 and section 5 have been satisfied.

4.3 Lotting. A production lot shall consist of the flocked grid screen manufactured by one producer at one plant, under essentially the same manufacturing conditions, using the same materials and equipment, and offered for acceptance at one time.

4.4 Sampling

4.4.1 For tests. Sampling shall be conducted in accordance with Table II. A sample 8 inches by 8 feet shall be taken which is not less than 2 feet from the lead end and not less than 3 inches in from the edge of the screen. No two samples shall be taken from any one roll or sheet. Specimens shall be cut from the samples in accordance with table III for the tests that are specified in 4.6.

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Table II. Sampling

Lot size (square yards)	Samples
Up to 200	1
200 to 1000	2
1000 to 10,000	3
Above 10,000	4

Table III. Test specimen size and number of tests

Test	Test Method	Specimen size (inches)	Number of tests
Porosity	4.6.1	7 x 7	3
Adhesion	4.6.2	7 x 7	1
Water immersion	4.6.3	7 x 7	1
Weathering	4.6.4	7 x 7	1
Low temperature	4.6.5	7 x 7	1
Mildew resistance	4.6.6	7 x 7	3
Blocking	4.6.7	8 x 8	1

4.4.2 Preparation for delivery. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.4.3 Plugged holes and edge. Each sheet or roll submitted shall be examined as specified in 4.5.2 and 4.5.3

4.5 Examination.

4.5.1 Preparation for delivery. The samples shall be examined in accordance with the classification of defects and with Standard MIL-STD-105.

4.5.2 Plugged holes. Place a 1-3/4 inch inside-diameter ring on a section of the flocked grid scrren at random and place an adequate light source under the section and count the plugged holes inside the ring. Foreach square yard, one examination shall be made.

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4.5.3 Edge. Measure the flock-free edges of the screen in three places on each side of the flocked grid screen for conformance with 3.2.4.1.

4.5.4 Classification of defects.

4.5.4.1 Preparation for delivery (see section 5)

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Containers incorrect
102	Interleaving paper missing from between flat sheets or layers on rolled material
103	Flat sheets compressed within container
104	Rolls unevenly or tightly wound
105	Outer wrap of rolled material not sealed as specified
106	Containers closure incorrect
107	Containers or wrapped rolls damaged
108	Marking incorrect, missing, or illegible
Minor:	AQL 10.0 percent defective
201	Initial wrap of rolled material missing or not secured as specified
202	Fiberboard circular pads missing from ends of wrapped wraps
203	Quantity of flatsheets per container not uniform
204	Length of material in roll not uniform

4.6 Tests. The following tests shall be conducted under the conditions specified in section 7 of Test Method Std. No. 141.

4.6.1 Air permeability. The air permeability of the flocked grid screen shall be determined in accordance with method 5450 of Specification CCC-T-191, except that the pressure drop measured on the inclined manometer shall be 0.2 inch. The specimens used for this test will also be used for 4.6.1.1.

4.6.1.1 Wet treating. In order to determine the wet air permeability, immerse the specimen from 4.6.1 in distilled water at room temperature for at least 30 seconds. Drain specimen of excess water for 15 seconds and then proceed, within 30 seconds, as in 4.6.1, over the same area.

4.6.2 Adhesion. Adhesion of the flock to the base screen shall be determined in accordance with method 6303 of Test Method Std. No. 141. The scratch tool with a weight of 1200 grams shall be moved across the flock-covered side of the screen for at least 4 inches. The path of the tool shall be examined under a 20-power magnification for flock-free base screen to meet the requirement of 3.2.2.

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4.6.3 Water immersion. The test specimen shall be immersed in boiling water for at least one hour. Remove the specimen and dry it for 24 hours at ambient conditions. Then test with scratch tool as specified in 4.6.2 to meet 3.2.2, except that the weight shall be 600 grams.

4.6.4 Weathering. The test specimen shall be weathered for 48 + 2 hours, in accordance with method 5804 of Specification CCC-T-191, and then tested with scratch tool, as specified in 4.6.2 to meet 3.2.2, except that the weight shall be 600 grams.

4.6.5 Low temperature. Place the test specimen into a cold box at -25°F and hold there for 24 hours. Test the specimen at -25°F with scratch tool as specified in 4.6.2 to meet 3.2.2, except that the weight shall be 600 grams.

4.6.6 Mildew resistance. Test the specimen as specified in method 5751 of Specification CCC-T-191.

4.6.7 Blocking. The specimen shall be tested in accordance with method 5872 of Specification CCC-T-191, except that it shall be exposed for six hours at 165°F and a pressure of 3 pounds per square inch to meet 3.2.3.

4.7 Acceptance/rejection criteria. RFailure to meet the requirements of section 3 shall result in rejection of the lot of flocked grid screens represented.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. The flocked grid screen shall be furnished in either flat sheets or in rolls as specified (see 6.2).

5.1.1 Flat sheets. Uniform quantities of flat sheets of flocked grid screen shall be packed in a W6c, style RSC, fiberboard box, conforming to Specification PPP-B-636. A sheet of paper, which is compatible with the material, shall be placed between each sheet of the material. Care shall be excercised to avoid compressing the contents of the box. The box shall be closed by taping all seams, including the manufacturer's joint, with minimum 3-inch wide, pressure-sensitive tape conforming to Specification PPP-T-76, extending the tape over all corners and edges at least three inches.

5.1.2 Rolls. The flocked grid screen shall be loosely and evenly wound on a hollow core constructed of fiberboard or other suitable material, with a continuous sheet of paper, compatible with the material, placed between layers of the material on the roll. When practicable, rolls shall contain uniform lengths of material. Individual rolls of the material shall have an initial wrap of grade B, 40-pound basis weight, kraft wrapping paper, conforming to Specification UU-P-268 secured in place with gummed tape conforming to Specification UU-T-111. Circular fiberboard pads, fabricated from material conforming to type I, class 1 of Specification PPP-B-636 shall be secured to each end of the

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roll, using the above-cited gummed tape. Each roll shall then be overwrapped with type C-1 or C-2 waterproof barrier material conforming to Specification UU-P-271 sealing all folds, laps, and seams in place with water-resistant adhesive conforming to Specification MIL-A-140.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers and rolls shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The flocked grid screen is intended for use in the inlet valve assembly of the field protective mask M17.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether rolls or sheets are required.

(c) Size of rolls or sheets.

6.3 Adhesive. A suitable adhesive has been found to be black deck enamel made by Kyanize Paints, Inc.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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